INVENTION EXHIBITION

Smart Bicycle Tail Light System

TAIWAN

PROJECT WEBSITE

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Physical demonstration model

THE DESIGN GOAL

Help people behind a bicycle recognize turns, braking and emergency warnings.



Rear light installed on a bicycle

INPUT	BLE LINK	PROCESS	OUTPUT
micro:bit V2	**Native services**	**XIAO ESP32-C6**	**8 LEDs + buzzer**
Buttons, touch logo, acceleration	Button, Accelerometer, Event Services	Calibration, filtering, state logic	WS2812B: D1 / Buzzer: D3 Software brightness cap

Native micro:bit BLE services; no UART. 400 mAh lithium battery.
Calibration remembers posture; the forward sensor axis and sign must still be set.

SIX CORE FEATURES

01 Riding and parking

Soft red with two green center LEDs while riding. Steady red after a configured period at rest.

02 Braking alerts

Startup calibration, moving averages and slow slope-baseline tracking process acceleration. A trigger flashes red LEDs and sounds the buzzer.

03 Turn signals

A: left. B: right. Four LEDs on the turning side show an amber sequence with sound. The other side keeps its lighting. Auto-cancel or timeout ends the signal.

04 Hazard lights

Short-press A+B to turn synchronized amber warning lights on or off.

05 SOS warning

Alternating red and white light with an alarm. Hold A+B continuously for 3 seconds to clear SOS.

06 Power-saving standby

After prolonged inactivity, the lights turn off. Movement restores them. The whole system stays powered.

OPERATING MODES

1 Demonstration mode

Upright calibration, sensitive braking and a sustained-tilt SOS trigger for the model.

2 Riding mode

Different braking, impact and tilt parameters aim to reduce unwanted triggers. Validation is pending.

Switch with the gold logo or the phone webpage.

OFFLINE PHONE SETTINGS

About 49 parameters: sensitivity, timing, filtering, tilt angle, lights, sound and standby.

Local Wi-Fi webpage.
No app or internet required.

TESTING AND RECORDS

1-100 rounds, default 30

Preparation and observation periods log triggers and acceleration. Manually confirm actual braking.

Exclude unperformed or unconfirmed rounds from the braking detection-rate denominator.

Verify actual light output by eye or video.

EVIDENCE AND CURRENT LIMITS

Braking detection rate = (confirmed braking rounds with a trigger / all confirmed braking rounds) x 100%.
CSV exports: trial results, acceleration, settings and summary. Offline reports compare outcomes and thresholds.
Compilation and computer-side checks completed. Detection rate, false-trigger rate and runtime remain unmeasured.
SOS is local only. No GPS or remote rescue alerts. No software battery/charge-state sensing. No road-use certification.